

Theory Week 6

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Mechanics 3 CAB G59
WhatsApp group



- Find all doc's on my website (link in WhatsApp bio)
- remember to do the STICK
- Study Center

6	21/10/24	angular momentum balance of rigid bodies	3.2.2
	23/10/24	moment of inertia tensor, Steiner's theorem	3.2.3
exercise 6: rigid-body kinematics & kinetics			

AMB

angular momentum:

$$\mathbf{H}_B = \lim_{\substack{N \rightarrow \infty \\ \Delta m_i \rightarrow 0}} \sum_{i=1}^n \mathbf{r}_{BP_i} \times (\Delta m_i \mathbf{v}_i) = \int_B (\mathbf{r} - \mathbf{r}_B) \times \mathbf{v} dm = \int_B (\mathbf{r} - \mathbf{r}_B) \times \rho \mathbf{v} dV. \quad (3.81)$$

balance of angular momentum with respect to an arbitrary point B:

$$\dot{\mathbf{M}}_B = \mathbf{H}_B + \mathbf{v}_B \times \mathbf{P}$$

angular momentum with respect to a point B on body B:

$$\mathbf{H}_B = \mathbf{I}_B \boldsymbol{\omega} + M(\mathbf{r}_{CM} - \mathbf{r}_B) \times \mathbf{v}_B$$

moment of inertia tensor (B ∈ B serves as coordinate origin):

$$[\mathbf{I}_B] = \int_B \begin{pmatrix} x_2^2 + x_3^2 & -x_1 x_2 & -x_1 x_3 \\ -x_1 x_2 & x_1^2 + x_3^2 & -x_2 x_3 \\ -x_1 x_3 & -x_2 x_3 & x_1^2 + x_2^2 \end{pmatrix} \rho dV$$

parallel axes theorem (Steiner's theorem) with $\Delta \mathbf{x} = \mathbf{r}_B - \mathbf{r}_{CM}$:

$$[\mathbf{I}_B] = [\mathbf{I}_{CM}] + M \begin{pmatrix} (\Delta x_2)^2 + (\Delta x_3)^2 & -\Delta x_1 \Delta x_2 & -\Delta x_1 \Delta x_3 \\ -\Delta x_1 \Delta x_2 & (\Delta x_1)^2 + (\Delta x_3)^2 & -\Delta x_2 \Delta x_3 \\ -\Delta x_1 \Delta x_3 & -\Delta x_2 \Delta x_3 & (\Delta x_1)^2 + (\Delta x_2)^2 \end{pmatrix}$$

If you choose B to simplify maths (& mistakes)

$$\hookrightarrow B = \text{CoM} \quad (\mathbf{r}_{CM} - \mathbf{r}_B = 0 \text{ \& } \mathbf{v}_B \parallel \mathbf{P})$$

$$\hookrightarrow B = \text{fixed} \quad (\mathbf{v}_B = 0)$$

$$\hookrightarrow \dot{\mathbf{I}}_B = 0$$

balance of angular momentum for B ∈ B if B = CM or $\mathbf{v}_B = 0$ and if $\dot{\mathbf{I}}_B = 0$:

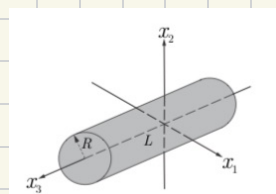
$$M_B = I_B \dot{\omega} \quad \xrightarrow{\text{in 2D}} \quad M_B = I_B \dot{\omega} \quad \text{with} \quad I_B = I_{CM} + M(\Delta x)^2$$

(sidenote: if body = massless, $M=0$ & $I_B=0$)

$$\Rightarrow \underline{\underline{M_B = 0}}$$

Moment of inertia tensor examples

$$\square \text{ Cylinder: } \underline{\underline{I}}_{CM} = \begin{pmatrix} \frac{M}{12}(L^2 + 3R^2) & 0 & 0 \\ 0 & \frac{M}{12}(L^2 + 3R^2) & 0 \\ 0 & 0 & \frac{MR^2}{2} \end{pmatrix}$$



!!! If an axis is ⊥ to a plane of symmetry, then all off diagonal terms with i-subscripts vanish

□ centroidal moment of inertia (I_{CM} in 2D)

$$\textcircled{1} \text{ slender rod: } I_{CM} = \frac{ML^2}{12}$$

$$\textcircled{2} \text{ disc/cylinder: } I_{CM} = \frac{MR^2}{2}$$

Moment of inertia tensor

(a "Tensor" is a matrix, but basis dependent) → see appendix

moment of inertia tensor = $[I_B]_C = \int_B \begin{pmatrix} x_2^2 + x_3^2 & -x_1x_2 & -x_1x_3 \\ -x_1x_2 & x_1^2 + x_3^2 & -x_2x_3 \\ -x_1x_3 & -x_2x_3 & x_1^2 + x_2^2 \end{pmatrix} \rho dV.$

parallel axes
or
Steiner theorem

$$[I_B] = [I_{CM}] + M \begin{pmatrix} (\Delta x_2)^2 + (\Delta x_3)^2 & -\Delta x_1 \Delta x_2 & -\Delta x_1 \Delta x_3 \\ -\Delta x_1 \Delta x_2 & (\Delta x_1)^2 + (\Delta x_3)^2 & -\Delta x_2 \Delta x_3 \\ -\Delta x_1 \Delta x_3 & -\Delta x_2 \Delta x_3 & (\Delta x_1)^2 + (\Delta x_2)^2 \end{pmatrix}$$

[2D: $I_B = I_{CM} + M(ax)^2$]

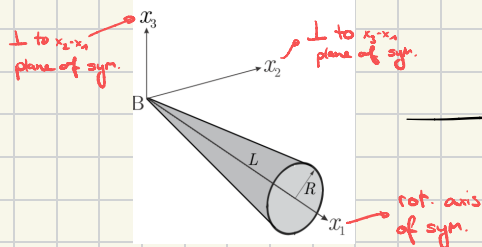
⇒ 3 conclusions: 1st: symmetric → $I_{ij} = I_{ji}$

2nd: additive → $I_B (\text{body 1} + \text{body 2}) = I_{B1} + I_{B2}$

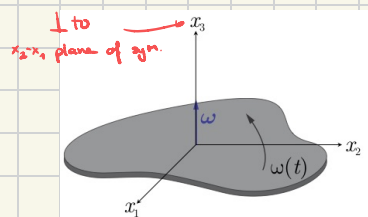
3rd: sym. axis → if x_i -axis is an axis of rot. sym., then all off-diagonal terms of I_B with i -subscript vanish

again → if x_i stands $\perp$ to sym. plane → all off-diag. i -subscript terms vanish

for example:



$$\begin{pmatrix} I_{11} & 0 & 0 \\ 0 & I_{22} & 0 \\ 0 & 0 & I_{33} \end{pmatrix}$$



$$\begin{pmatrix} I_{11} & I_{12} & 0 \\ I_{21} & I_{22} & 0 \\ 0 & 0 & I_{33} \end{pmatrix}$$

principle axes: (principal) coordinate system → I_B = diagonal

$$[I_B]_{\hat{C}} = \begin{pmatrix} I_{11} & 0 & 0 \\ 0 & I_{22} & 0 \\ 0 & 0 & I_{33} \end{pmatrix}$$

eigenvalues (pointing to the diagonal elements)

eigenbase (pointing to the matrix)

remember LinAlg: when Tensors are written in Eigenbasis, then they are diag. with Eigenvalues on diag.

↳ axes of rot. sym & axes $\perp$ to plane of sym. are always principle axes